

Lab-tut problems for Week 2

Aim of these lab-tut problems: By working on these problems, we understand how, in practice, how to, based on a continuous time model, obtain an approximate discrete time version. Then, based on the obtained discrete time model, we can easily implement simulations and predictions. We also consider kinematic models, related to the type of platform we will use on our projects.

In addition, by solving these problems you may re-activate your skills in programming, which is be convenient for solving the assignments in the course. The acquired knowledge and the implementations (source code) that you develop for answering these questions will be useful in subsequent projects in our course.

Question 1.

Given a pendulum system, whose nominal model is expressed by the following ODE,

$$\ddot{\varphi} = -9 \cdot \sin(\varphi) - 0.5 \cdot \dot{\varphi} + 2 \cdot u(t)$$

(in which $\dot{\varphi}$, $\ddot{\varphi}$ are expressed in radians, radians/sec and radians/sec², respectively, and the participating constants are expressed in units which are consistent with those of the variables.)

- 1.1) Obtain a valid state space model.
- 1.2) Propose a discrete time approximation for the model obtained in (1.1), assuming a regular sampling period $T=2$ milliseconds.
- 1.3) Exploiting the model from (1.2), implement a Matlab function for predicting, in just one step, $\mathbf{x}(t+T)$, based on knowing $\mathbf{x}(t)$ and $\mathbf{u}(t)$ (i.e., your Matlab function is executed being provided with those variables through the function input arguments).
- 1.4) Implement a small program, for simulating the pendulum, based on the discrete time model that you obtained in the previous item of this question.
- 1.5) Run a simulation, assuming the following conditions:
 - a) Initial angular position and initial angular rate $\varphi(t_0) = 110^\circ$; $\dot{\varphi}(t_0) = 0^\circ / \text{sec}$, respectively.
 - b) Input $u(t) = 0$, $\forall t$.

Simulate the system for an interval of time $0 \leq t \leq 20$ (t expressed in seconds)

Plot the simulated angular position and angular velocity, for that interval of simulation.

- 1.6) Repeat the simulation, assuming that the input is $u(t) = \begin{cases} 0 & t < 5 \\ 4 & t \geq 5 \end{cases}$

Question 2.

Suppose that you drive a car, for which you have the following approximate kinematic model when operated on flat horizontal terrain.

$$\frac{d}{dt} \begin{bmatrix} x(t) \\ y(t) \\ \phi(t) \end{bmatrix} = \begin{bmatrix} v(t) \cdot \cos(\phi(t)) \\ v(t) \cdot \sin(\phi(t)) \\ \frac{v(t)}{L} \cdot \tan(\beta(t)) \end{bmatrix}$$

a] What is the approximate heading rate that you would measure, if, when driving your car, you applied a steering angle of 20 degrees, when car's speed (longitudinal velocity, $v(t)$) were 20km/h?

b] What steering angle should you have applied for experiencing, at that time, a heading rate $\dot{\phi}(t_0) = -10^\circ / \text{sec}$?

Note: if you do not have a car, just choose the one you prefer, i.e. your friend's car, or assume you borrowed a tricycle from some kid you know (in such a case, better to assume a lower speed!).

Why do we need to know which vehicle are you driving?

Question 3.

What would occur if you, when driving a vehicle during an interval of time, kept the steering angle constant?

(Implement and perform a brief simulation, for a small car (e.g. having $L=1.5\text{m}$), and for a large one (e.g. $L=3\text{m}$), using a reasonable steering angle. Apply an approach similar to that used in Question 1.

What do you see from the plots? (from your simulations, no need to waste fuel/energy doing it with a real car!)

Note: you may consider a sampling period $T=15\text{ms}$ (why? To be discussed in lecture time).

Question 4.

Based on your program for question 3, perform a simulation in which during 5 seconds you move forward, at speed = 10m/s, having steering angle 10 degrees, then you stop for about 2 seconds, during which you change gear to reverse, and the steering angle to -10 degrees, and then, in no time (this is unfeasible, but we simulate it, anyway), you achieve a cruise speed of -3m/s. Plot the resulting trajectory in 2D. Does it make sense? Is the model valid for negative speeds?

Note: Alternatively, you may replace the aggressive speed changes by using continuous transitions. (optional in this question).

Questions/doubts: ask the lecturer, via Moodle or via email (j.guivant@unsw.edu.au), or consult your demonstrators during tutorial time.

The Lecturer will show his solution, during the lecture on week 2. Details about how to implement the processing will be shown and discussed.

Note: Substantial part of the code you implement for solving this lab-tutorial, may be used by you, later, when you solve assignment 1. Solving these questions, is a good investment of effort, due to many reasons.